

Heterogeneity Within Molecular Subtypes of Breast Cancer

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Running Head: Dynamic Molecular Subtypes in Breast Cancer

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Abstract:

Breast cancer is the quintessential example of how molecular characterization of tumor biology guides therapeutic decisions. From the discovery of the estrogen receptor to current clinical molecular profiles to evolving single cell analytics, the characterization and compartmentalization of breast cancer into divergent subtypes is clear. However, competing with this divergent model of breast cancer is the recognition of intratumoral heterogeneity, which acknowledges the possibility that multiple different subtypes exist within a single tumor. Intratumoral heterogeneity is driven by both intrinsic effects of the tumor cells themselves as well as extrinsic effects from the surrounding microenvironment. There is emerging evidence that these intratumoral molecular subtypes are not static; rather, plasticity between divergent subtypes is possible. Inter-conversion between seemingly different subtypes within a tumor drives tumor progression, metastases, and treatment resistance. Therapeutic strategies must therefore contend with changing phenotypes in an individual patient's tumor. Identifying targetable drivers of molecular heterogeneity may improve treatment durability and disease progression.

Introduction:

Historically, heterogeneity in breast cancer is understood in terms of varying expression of the Estrogen Receptor (ER), Progesterone Receptor (PR), and Human Epidermal Growth Factor Receptor 2 (HER2). Clinically these markers have come to define breast cancer, informing both prognosis and treatment paradigms. More recently, the Prediction Analysis of Microarray 50 (PAM50), which classifies tumors into five intrinsic subtypes (Luminal A, Luminal B, HER2-enriched, Basal, Normal-like) based on mRNA expression of a panel of 50 genes, has further advanced the characterization of breast cancer(1, 2). Interestingly, the molecular signature across breast cancer subtypes does not appear to represent a permanent state but rather a provisional one, capable of inter-conversion(3-5). This example of intratumoral heterogeneity (ITH) is due to two cooperative principles: clonal heterogeneity and cell state heterogeneity. Clonal heterogeneity is defined by phenotypic variance within a tumor, marked by both spatial and temporal factors(6, 7). Clonal evolution of a dominant or resistant clone is a potential mechanism for malignant spread and treatment resistance(8, 9). Both the malignant epithelium and the tumor microenvironment have been shown to play a role in the induction and perpetuation of clonal heterogeneity(6, 10). Cell state heterogeneity helps to explain how different cell states can co-exist in a single tumor, some with stem properties, some with progenitor like properties, and some with differentiated properties(11). The existence of a spectrum of cells states has clear clinical applications, as targeted therapeutics must address this heterogeneity to prevent resistance. Taken together, clonal heterogeneity and cell state heterogeneity account for the observation that new and distinct breast cancer subtypes can evolve from an ostensibly uniform tumor. Single cell analysis of breast cancer tumors promises

new insights into the drivers of ITH(12, 13). This review seeks to highlight advances in the molecular underpinnings of the existence of multiple different breast cancer subtypes in a single tumor and how this work informs the clinical management of disease.

Main Body:

Compartmentalization of Clinical Breast Cancer Based on Molecular Signatures

Breast cancer is the first human cancer whose treatment algorithm was driven by complex molecular characterization of patient specific tumors. Anti-estrogen therapy was one of the first examples of precision medicine, taking into account the molecular signature of an individual patient (ie. overexpression of the ER), to effectively treat a subset of patients with breast cancer(14, 15). The field of individualized treatment in breast cancer was advanced with the discovery of HER2 overexpression in ~25% of breast cancers and subsequent effective anti-HER2 therapy (e.g., trastuzumab, pertuzumab)(16-18). ER, PR, and HER2 form the backbone of clinical decision making to tailor therapy in patients with breast cancer and improve oncologic outcomes. These hallmarks, combined with newer biomarkers including Breast Cancer Gene 1/2 (BRCA 1/2) and Programmed Cell Death 1 (PD-1), among others, help to guide the modern management of breast cancer.

Though this system offers great insight into individual tumor susceptibilities, further characterization and compartmentalization was necessary to reconcile observed clinical heterogeneity (Figure 1). Complete molecular classification of breast cancer was first made possible with the PAM50 classification system which classifies breast cancer tumors into five distinct subtypes: Luminal A, Luminal B, HER2-enriched, Basal, and Normal-like(1, 2, 19). Additional work has further refined this classification schema. Prat *et al.* described the claudin-low subtype, which has both unique molecular signatures and is phenotypically different from the five original subtypes(20). The claudin-low subtype is defined by lack of luminal differentiation markers, and increased epithelial-mesenchymal transition (EMT), immune

response, and cancer stem cell (CSC) signatures. This subtype commonly represents triple negative breast cancers (TNBCs) with a poor prognosis. Subtyping breast cancer has been shown to be independently prognostic in multiple different patient cohorts and a more complete classification schema has been developed to predict clinical response(2, 21).

A separate classification schema was put forward by Curtis *et al.*, who identified 10 integrated clusters from 2,000 breast tumors using copy number and gene expression profiles to define their cohorts(22). Each subtype had distinct copy number aberrations, consistent in both their discovery and validation cohort. Their integrated cluster system stratified patients by prognosis with each individual cluster having distinct clinical outcomes. They identified a cluster of ER⁺ tumors with poor survival and mutations in several potential oncogenic drivers such as Cyclin D1 (*CCDN1*), EMSY Transcriptional Repressor (*EMSY*), and P21 Activated Kinase 1 (*PAK1*). Longitudinal patient follow-up, over a median of 14 years, led to the identification of four clusters of patients with traditionally indolent ER⁺HER2⁻ breast cancer who experienced recurrent disease(23). Whereas a subgroup of TNBC patients, who classically develop significantly more aggressive disease, remained relapse free. These two examples highlight how a comprehensive understanding of the molecular classification of breast cancer translates directly to improved tumor-specific patient care and outcomes. The most well-studied clinical subtype of breast cancer in terms of molecular heterogeneity and subclassification is TNBC. Lehmann *et al.* first demonstrated complex heterogeneity utilizing gene expression analysis of 587 TNBC cases(24). They found six subtypes of TNBC, termed Basal-like 1 and 2 (BL1 and BL2), Immunomodulatory (IM), Mesenchymal (M), Mesenchymal Stem-Like (MSL), and Luminal Androgen Receptors (LAR) which they further refined to four subtypes: BL-1, BL-2, M,

and LAR (Figure 1). Each subtype has biologically unique, subtype specific driver pathways(25). When analyzed for prognostic ability against five large cohorts of breast cancer patients, they found that significantly more BL1 patients achieved a pathologic complete response as compared to the less chemotherapy responsive BL2 and LAR subtypes. Berache *et al.* further highlighted the genomic heterogeneity between these subtypes establishing that while the BL1 subtype is characterized by genomic instability, LAR tumors are associated with high mutational burden/Phosphatidylinositol 3-Kinase (*PI3KCA*) mutations, and the M subtype is associated with angiogenesis(26). Additional classification systems have also found four distinct subtypes with prognostic implication(27, 28). Notably, several potential therapeutic targets have been identified among these four subtypes: 1) Luminal Androgen Receptor – marked by androgen receptor, cell surface protein Mucin 1 (*MUC1*); 2) Mesenchymal – marked by growth factors (Cluster of Differentiation 117 (*CD117*), PDGF receptor A); 3) basal-like immune-suppressed – marked by immune suppressive molecule V-Set Domain-Containing T-Cell Activation Inhibitor 1 (*VTCN1*); 4) basal-like immune-activated – marked by Stat signaling molecules, cytokines(28).

Single cell RNA-sequencing was recently utilized to explore this heterogeneity within six patients with primary, non-metastatic TNBC. All tumors were of similar grade and analyzed prior to initiation of any treatment. These investigators found that TNBCs are composed of tumor specific clusters that correspond to subpopulations defined by large clonal copy number variations, consistent with phenotypes seen in other tumor types(29-31). Distinct subgroups shared between different patients led to the identification of a single subpopulation which was associated with treatment resistance(31). This genetic signature of therapeutic resistance was characterized by glycosphingolipid metabolism and lysosomal turnover affecting the cytokine

pathways involved in the innate immune response. This study highlights the profound intratumoral heterogeneity within TNBC while also unmasking a conserved mechanism of resistance across TNBC patients.

Dynamic Conversion Between Disparate Molecular Subtypes

Added to the complexity of ITH is the possibility for dynamic interconversion between distinct molecular subtypes. Priedigkeit *et al.* studied 20 patients with primary tumors and resected brain metastasis(3). PAM50 subtype concordance between the primary tumor and resulting metastatic disease occurred in 17 of 20 patients. However, there were expression changes in a number of other targetable genes including fibroblast growth factor 4 (*FGFR4*), Fms Related Receptor Tyrosine Kinase 1 (*FLT1*), Aurora Kinase A (*AURKA*), and *HER2* (32-35). Of the 13 patients with *HER2*⁻ primary tumors, 3 patients had *HER2*⁺ brain metastasis, demonstrating not only the dynamic conversion between subtypes but also highlighting the potential for targetable therapy in metastatic disease. These findings were validated in two additional independent cohorts, finding enrichment in *HER2* signaling in 11-22.2% of patients with brain metastasis despite *HER2*⁻ primary tumors. While the exact mechanism responsible for switching from primary *HER2*⁻ tumors to brain *HER2*⁺ metastasis is unknown, one attractive possibility is that the microenvironment present in the brain provides a niche that favors *HER2* signaling. Notably, dynamic conversion between subtypes is not isolated to brain metastases. A systematic review comparing primary tumor and metastatic progeny found frequent conversion in ER, PR, and *HER2* expression(36). Loss of function, noted as a positive to negative conversion in receptor expression from primary tumors to paired metastasis, occurred in 22.5%, 49.4%,

and 21.3%, respectively. Gain of function changes in these markers were seen in 21.5%, 15.9%, and 9.5% of patients, respectively. These conversions were dependent on the metastatic tumor environment, with ER conversion rates higher in bone and the central nervous system and lower in the liver, again pointing to a potential metastatic niche. However, the exact mechanisms for this change in expression remains to be elucidated. Three potential explanations are selection for a pre-existing clone that may have been masked by signatures of the bulk tumor, a change in the molecular expression of ER/PR/HER2, or a combination of both potentialities acting in concert. Interestingly, Garcia-Recio *et al.* found a group of luminal A tumors that give rise to HER2⁺ metastasis but remain non-responsive to targeted HER2 treatment(4). In their model, standard anti-estrogen therapy led to resistance and induction of the HER2⁺ phenotype that then reverted back to the more indolent luminal A upon inhibition of FGFR4. Clearly, dynamic conversion between breast cancer subtypes is a potential mechanism for treatment resistance and metastatic disease. In-depth characterization of the molecularly distinct cells within the primary tumor and the mechanism of plasticity may shed new light on how to best target future therapeutics.

Cell State Heterogeneity as a Driver of Intratumoral Heterogeneity

Hierarchical Organization Within the Normal Mammary Tissue and in Breast Cancer

The mammary differentiation hierarchy is complex. At the apex of the hierarchy is the mammary stem cell, capable of regeneration or differentiation in response to physiologic stimuli(37, 38). There exists a spectrum of lineage specific basal, hormone-sensing, and alveolar cells(37-39).

The proposed cell state model, utilizing the framework of the normal mammary hierarchy, helps to explain the potential existence of multiple cell states within a single tumor(11, 40-43). Yeo *et al.* proposed that within the spectrum of tumor cell states there is inter-conversion and therefore plasticity between breast cancer subtypes(43). In a follow-up study, they performed single cell RNA sequencing on three separate murine models of breast cancer (Mouse Mammary Tumor Virus (MMTV) - Polyoma Middle Tumor-Antigen (PyMT), MMTV-Neu, and BRCA1-null)(11). Within each tumor they found evidence of multiple breast cancer subtypes, further demonstrating need for cellular resolution in identifying combinatorial therapeutic strategies that account for the varying subtypes present. Risom *et al.* advanced this line of thought in the basal-like subtype, finding a population of cells resistant to therapy, termed drug tolerant persister (DTP) cells, which are driven by distinct cell state transitions due to remodeling of open chromatin architecture, rather than clonal evolution(44). Cell state plasticity is potentially targetable, as evidenced by the observation that treatment with PI3K/Mechanistic Target of Rapamycin (mTOR) inhibitors prevented DTP cell production when used in combination with Bromodomain and Extra-Terminal Motif (BET) inhibitors in the HCC1143 cell line. *In vivo* xenograft models found tumor regression when mTOR inhibitors and BET inhibitors were used in combination(44). These models demonstrate a complex tumor organization, with multiple cell states capable of inducing heterogeneity within the breast tumor.

Implications of Multiple, Distinct Breast Cancer Stem Cells

The CSC hypothesis posits that a small number of pluripotent cells, capable of regeneration and differentiation, drive tumor proliferation, invasion, metastases, resistance to drug therapy, and recurrent disease. While the concept has been around for approximately half a century, it has come to the forefront over that past 20 years. There are numerous well written reviews on CSCs, highlighting their role in disease progression and implications for treatment resistance(45-48). Within the breast cancer literature multiple markers of CSCs have been described including cell surface molecules such as CD44⁺/CD24^{-/low} (49), Aldehyde Dehydrogenase (ALDH)⁺ (50), CD133⁺ (51), and Protein C Receptor (PROCR)⁺ (52). One potential explanation for the heterogeneity of CSC markers in mammary tumors is their niche specific behaviors, which may confound studies attempting to identify one over-arching CSC(46). Another possible explanation is that this heterogeneity represents the same diversity seen in normal mammary progenitor cells due to spatial and temporal factors, consistent with the mesenchymal-like and epithelial-like CSCs(43). Regardless, this evidence suggests that multiple cell states may confer a CSC phenotype, potentially driving intratumoral heterogeneity.

Heterogeneity at the CSC level may explain some of the many different pathways shown to regulate CSC-like behavior. For example, Tam *et al.* examined the effects of Fos-related antigen 1 (FRA1) as a downstream effector of PKC α in driving formation of CSCs(53). They found that inhibition of either FRA1 or PKC α prevented tumorigenesis in basal-like breast cancer model. CSC function is also dependent on plasticity between EMT and mesenchymal-epithelial transitions (MET), allowing this subpopulation to promote metastasis, invasion, and treatment resistance(40). Autophagy is another cellular process key to CSC survival. Work from our lab identified two distinct breast CSC populations, termed ALDH⁺ and CD29^{hi}CD61⁺, in a murine

model(54). We found that while both CSC populations had tumor initiating properties, the more basal-like CD29^{hi}CD61⁺ breast CSCs exhibited increased EMT signatures, while the ALDH⁺ breast CSCs were closely associated with luminal progenitors. Inhibition of autophagy through deletion of an essential autophagy gene, Focal Adhesion Kinase Interacting Protein of 200 kD (FIP200), prevented the tumor initiating properties of both CSC populations, via two different pathways (Stat3 and Transforming Growth Factor β (TGF β), respectively). In vivo inhibition of tumor growth was accomplished by targeting both CSC populations at once, with Stat3 and TGF β chemical inhibition. In another study, Ithimakin *et al.* showed that luminal CSCs were dependent on a HER2 signaling node, present even in the absence of HER2 amplification(55). They showed that administration of anti-HER2 therapy prevented tumor engraftment, not tumor growth, in mouse xenograft models of luminal breast cancer, pointing towards selective inhibition of the CSC phenotype. This observation is particularly important as clinical HER2 status is determined by immunohistochemistry +/- fluorescent in situ hybridization; both methods are operator dependent and therefore subject to wide variability and may ultimately result in missing a CSC subpopulation responsive to anti-HER2 therapy(56). Targeting CSC behavior by accounting for the multiple distinct signaling nodes controlling their phenotype is a potential way to combat ITH.

Epithelial-Mesenchymal Transitions Driving Cell State Heterogeneity

EMT is characterized along a spectrum, wherein cells transition from a differentiated epithelial state typified by cell-cell adhesion and lack of motility, to a mesenchymal state typified by motility and invasiveness. Cells traverse this continuum of reversible states, very

often undergoing partial EMT where both epithelial and mesenchymal features are present. The concept of reversibility and partial EMT reconciles the observation that metastatic tumor cells largely present with an epithelial phenotype without clear features consistent with a mesenchymal transition. Importantly, EMT signatures have been associated with certain breast cancer subtypes (ie. claudin-low, basal) and breast cancer stem cell plasticity(6, 20, 57). The high EMT signatures associated with both the basal and claudin-low subtypes could partially explain the aggressive nature of these subtypes, as the increased motility and invasive features of a mesenchymal state can contribute to this aggressive phenotype(57). Further, plasticity between EMT and MET states in response to specific stimuli is thought to contribute to observed heterogeneity in TNBC and drive metastasis within this subtype(58).

Tumor-Microenvironmental Factors and their Role in Defining Tumor Heterogeneity

Cancer-Associated Fibroblasts

Cancer-associated fibroblasts (CAFs) are a major component of the tumor microenvironment, promoting cancer progression through creation of pro-tumorigenic extracellular matrix (ECM) structures, generation of growth factors, regulating tumor immunity, modulating chemoresistance, and promoting metastases(59, 60). Roswall *et al.* found that PDGF-CC was preferentially expressed in the basal-like molecular subtype of breast tumors and that paracrine crosstalk between CAFs and cancer cells expressing PDGF-CC determined the molecular subtype of the tumor, luminal versus basal(61). They were then able to sensitize ER⁺ tumors to anti-estrogen treatment by blocking PDGF-CC signaling via both genetic and chemical inhibition. Another study demonstrated that CAFs provide a supportive niche for

chemoresistant CSCs by increasing hedgehog mediated upregulation of FGF5 expression and fibrillar collagen(62). Inhibition of the CAFs in patient derived xenografts led to a decrease in the CSC phenotype and rescued response to docetaxel. Targeting CAF driven ITH, via inhibition of pathways such as PDGF-CC and FGF5, represents a novel mechanism to prevent tumor cell plasticity and resistance.

Heterogeneity among CAFs themselves has also been shown to lead to a treatment resistant phenotype. Four distinct classes of CAFs are differentially associated in various molecular subtypes of breast cancer, with the CAF-S1 and CAF-S4 preferentially found in TNBC(10). The CAF-S1 subtype is described to induce an immunosuppressive environment by upregulating CD25^{High}FOXP3^{High} (Foxhead Box P3) T cells and enhancing T regulatory (T_{reg}) capacity. Further work has demonstrated eight discrete clusters within the previously described CAF-S1 class(63). The researchers demonstrated that within these clusters a positive feedback loop existed between distinct clusters and T_{reg} cells, driving immunoresistance. Su *et al.* demonstrated that a new subset of CAFs, defined as CD10⁺GPR77⁺ (G Protein-Coupled Receptor 77), maintained cancer stemness and promoted resistance to cytotoxic chemotherapy(64). Through antibody targeted inhibition of GPR77 signaling, they were able to rescue response to docetaxel *in vivo*. CAF heterogeneity and its contribution to tumoral resistance is an important area for future study.

Infiltrating Immune Cells and Immunotherapy

Leveraging the immune system in the fight against breast cancer became a reality with the KEYNOTE-355 and 522 clinical trials, which demonstrated that the addition of checkpoint

inhibitors (i.e., pembrolizumab) to standard chemotherapy improved clinical outcomes in patients with metastatic as well as early stage TNBC(65, 66). However, response to immunotherapy has not been uniform in breast cancer, even when limited to patients with TNBC, pointing to clinically relevant heterogeneity. To better understand this phenomenon, Jézéquel *et al.* utilized gene-expression profiles to conduct unsupervised clustering in 107 TNBC patients(67). They discovered three subtypes of TNBC: 1) Luminal Androgen Receptor 2) Basal-like with low immune response and high M2-like macrophages 3) Basal-like with high immune response and low M2-like macrophages. Patients with the molecular signature associated with high immune response and low M2-like macrophages had improved event-free survival compared to patients with high M2-like macrophage signatures. Clearly, a deeper understanding of tumor heterogeneity and the crosstalk between specific cancer cell types and the immune microenvironment will help guide tumor-specific therapy in the future.

Single cell analysis has also been utilized in the study of the immune system in breast cancer. Chung *et al.* utilized single-cell transcriptomics to study immune cells in 11 patients with breast cancer; importantly, their cohort of patients included TNBC tumors as well as luminal A, luminal B, and HER2⁺ tumors(68). They found the dominant population of non-carcinoma cell signatures were immune cells, which they divided into three immune cell signatures corresponding to T and B lymphocytes as well as macrophages. The dominant signature in their cohort was immunosuppressive, with T cells demonstrating an exhausted/regulatory phenotype and macrophage signatures consistent with an immunosuppressive M2 phenotype. Wagner *et al.* generated a comprehensive atlas of the tumor and immune microenvironment for human breast cancer utilizing single-cell proteomics(69). In their large-scale study, they

describe a complex tumor ecosystem, with wide heterogeneity both among the tumor cells and within the immune microenvironment, working in concert to govern disease progression. Specifically, they found that PD-L1⁺ (Programmed Death-Ligand 1) expressing tumor associated macrophages and exhausted T cells accounted for patient risk stratification in both ER⁺ and ER⁻ tumors.

Clonal Evolution and Cooperativity as a Cause of Heterogeneity

Genetic factors also play an integral role in the development of ITH. Genetic contributions are clearly apparent in branching evolution, with both driver and passenger mutations influencing the fitness of a given phenotype within their complex microenvironment(8). Temporal heterogeneity defined by clonal evolution was seen in a longitudinal study of one patient with metastatic ER⁺/HER2⁺ breast cancer who received targeted therapy over three years(70). They analyzed the response to therapy by assessing eight tumor biopsies and nine plasma samples in real time and found that circulating tumor DNA could predict clonal evolution. Changes in sub-clonal mutations directly correlated with treatment response at different sites of metastatic disease.

An important and well-studied pathway in defining clonal evolution is the Wnt signaling pathway. Cleary *et al.* first demonstrated that Wnt signaling induces mixed-lineage tumors, that conform to both an expected hierarchical organization from a single progenitor and a multi-clonal tumor cell composition giving rise to distinct basal and luminal subclones(71). Follow up work found that Receptor Tyrosine Kinase Like Orphan Receptor 2 (Ror2) knockdown led to augmented Wnt/ β -catenin signaling, yielding divergent gene expression and distinct

phenotypes among different basal-like tumors(72). This work provides a context for the Ror2 divergent spatiotemporal function in regulating cellular heterogeneity in tumor progression through altered Wnt signaling. The role of the Wnt signaling pathway in driving phenotypic determination of breast cancer cells offers a potential target to prevent clonal evolution and the formation of ITH.

Cooperative intratumoral heterogeneity has also been seen between mesenchymal-like cells and tumor-initiating cells (TICs) in p53 null mouse models of basal-like breast cancer(73). Abrogation of the feedback between the mesenchymal-like cells and TICs via knockdown of ligands on the mesenchymal-like cells or receptors on the tumor-initiating cells prevented tumorigenesis. Further analysis of a subpopulation of mesenchymal cells, denoted as CD29^{High}CD24^{Low} cells, revealed a gene expression signature consistent with the claudin-low subtype of breast cancer, which is known to have increased EMT features, potentially driving ITH in these tumors. This study argues for a cooperative role between distinct subpopulations within the same tumor and notes a new potential target to combat tumorigenesis: the inter-relationship between different subtypes.

Clinical Intratumoral Heterogeneity and Resulting Implications

Clinically, breast cancer ITH can be expressed in terms of both spatial and temporal heterogeneity. Different areas of the same tumor can express ER, PR, and HER2 at different levels by immunohistochemistry(74). Consistent with this observation, next generation sequencing of 21 primary breast cancers with different clinical phenotypes (ER/PR/HER2), found distinct subclones with different mutational profiles within each tumor(75, 76). Spatial

heterogeneity was also studied utilizing imaging mass spectrometry to perform single cell analysis in 352 breast cancer patients who encompassed all clinical subtypes and pathologic grades(77). These authors showed that phenotypic heterogeneity in breast cancer tumors was spatially localized to distinct regions and patients with high levels of phenotypic spatial heterogeneity had poorer outcomes. Mechanisms driving these distinct regions of heterogeneity require further study.(78)

Temporal evolution leading to ITH is best understood when comparing primary tumors to their metastatic descendants. Several studies have demonstrated almost complete divergence between primary tumors and metastases(74). Treatment strategies to combat ITH must be robust to combat the resistance ultimately derived from this divergence. Proposed solutions include combination therapy, exploiting passenger mutations, eradicating the “lethal” clone, adaptive therapy, targeting the tumor microenvironment, and immunotherapy(74, 79, 80). Targeting CSCs, one paradigm to account for ITH and tumor plasticity, is also an attractive therapeutic target. In their review, Brooks *et al.* discuss combining traditional cytotoxic chemotherapy against bulk tumor with targeted CSC therapy to improve response(81).

Intratumoral Heterogeneity and its Contribution to Metastasis

Dynamic heterogeneity has the ability to promote metastasis and govern niche behavior in the metastatic microenvironment(82). Metastatic colonization is an inefficient process by nature, with only a minority of cancer cells able to survive in the harsh and foreign environment defined by the site of metastasis. Malanchi *et al.* demonstrated that breast CSCs are needed for metastatic colonization and that this colonization is also dependent on the stromal niche of the

metastatic site(83). Utilizing the MMTV-PyMT mouse breast cancer model, they found that tumor cells induce stromal periostin expression, which was essential for CSC maintenance. Inhibition of periostin signaling within the stroma prevented metastases, highlighting the cooperative relationship between metastatic tumor cells and the niche environment. Their work identified a role for direct targeting of the tumor microenvironment as a means to impede breast CSC driven metastases.

Genetic heterogeneity between primary tumor and the metastatic site is a well described phenomenon, seen in both treatment naïve and treatment resistant patients(84, 85). Lawson *et al.* used single-cell analysis of patient derived xenografts to identify metastatic cells in peripheral tissues and modeled the CSC paradigm in the metastatic setting(86). Their group recapitulated a hierarchical model of metastasis, where cells from tissues with low metastatic burden had tumor initiating abilities in transplantation experiments as well as the ability to differentiate into more mature luminal cells. Correspondingly, these cells from low metastatic burden tissue had increased expression of CSC, EMT, and pro-survival genes. In contrast, cells from tissues with high metastatic burden were more similar to primary tumor cells, with increased expression of luminal differentiation genes. This model is consistent with a CSC model for metastasis, where a stem-like cell initiates metastasis and differentiates into more mature cancer cells

Contrary to the single cell model for metastasis, Cheung *et al.* used multicolor lineage tracing to describe cell cluster metastasis, an established clinical observation of heterogeneous tumor cell clusters that are present throughout metastasis(87). In their study, cell cluster metastasis accounted for more than 90% of metastasis in the MMTV-PyMT mouse model.

Keratin 14 (K14)⁺ epithelial cells were necessary for this polyclonal tumor metastases model. Blocking the K14⁺ signaling pathway prevented metastasis despite the addition of known metastatic drivers. They describe a cooperative model between distinct breast cancer subtypes to achieve a common goal (i.e., metastatic dissemination).

Single-cell RNA sequencing was applied to TNBC patient derived xenografts, finding heterogeneity among primary tumors and micrometastatic disease, although micrometastatic disease contained a preserved intertumoral transcriptome, defined by upregulation of oxidative phosphorylation(88). Inhibition of oxidative phosphorylation attenuated metastatic spread to the lungs in a xenograft model of TNBC, without affecting primary tumor growth or engraftment, suggesting key differences between drivers of growth in the metastatic progeny and the primary tumor. Targeting metastatic progeny selectively offers an attractive option for treatment in a disease where metastasis is deadly.

Treatment Resistance Mediated through Tumoral Heterogeneity

Both inter and intratumoral heterogeneity are thought to mediate therapeutic resistance. Guarneri *et al.* examined rates of pathologic response among 107 patients with HER2⁺ breast cancer(89). They found that loss of HER2 expression was observed in more patients who underwent neoadjuvant chemotherapy alone, compared to patients who underwent chemotherapy in addition to treatment with targeted anti-HER2 agents. Further, loss of HER2 expression was associated with an increased rate of relapse, demonstrating dynamic conversion to a chemoresistant phenotype. In a large cohort of patients from Japan treated with neoadjuvant chemotherapy, the majority of whom received cytotoxic

chemotherapy regimens alone, 21.4% of initially HER2⁺ tumors were HER2⁻ post neoadjuvant treatment, further reinforcing dynamic conversion with resultant chemoresistance(90). The molecular mechanisms driving HER2 loss remain to be elucidated.

Tumors from patients with TNBC are distinctly heterogenous by nature. Studies in TNBC patients found chemoresistant subpopulations in response to chemotherapy(91). Notably, next-generation sequencing performed on matched samples from patients pre and post treatment detected genetic alterations in potentially actionable pathways in 90% of patients, including cell cycle alterations, PI3K/mTOR alterations, growth factor amplifications, Ras/MAPK alterations, and DNA repair genes. The inherently heterogenous nature of TNBC provides a more diverse cell population from which chemoresistance may be derived. Tumoral evolution can also lead to therapeutic resistance in a variety of clinically distinct breast cancers. Law *et al.* demonstrated that inhibiting Apolipoprotein B mRNA Editing Enzyme Catalytic Subunit 3B (APOBEC3B) dependent tumor evolution can promote hormonal resistance in models of ER⁺ breast cancer(92). Brady *et al.* found clonal evolution in four patients with metastatic ER⁺ breast cancer, utilizing both whole exome sequencing and single cell RNA sequencing(93). These authors observed that pre-existing resistant phenotypes, including enhanced mesenchymal/growth signals (promotes drug resistance) and decreased antigen presentation/TNF- α signaling (immunosuppressive), became dominant post-treatment. Using single cell DNA and RNA sequencing, Kim *et al.* also found clonal persistence vis-à-vis adaptive genomic selection and transcriptional reprogramming among resistant patients with TNBC(94). In the HER2 enriched subtype, patients treated with two independent anti-HER2 agents (lapatinib and trastuzumab), induced a low proliferative Luminal A phenotype(95). These results

corroborated *in vitro* models which also observed induction of the low proliferative Luminal A phenotype with dual anti-HER2 therapy, seen more commonly in hormone receptor positive, HER2 enriched disease compared to hormone receptor negative, HER2 enriched disease. The Luminal A phenotype was associated with an increased sensitivity to Cyclin Dependent Kinase (CDK) 4/6 inhibitors. Upon discontinuation of the anti-HER2 therapy *in vitro* or acquired resistance *in vivo*, the original HER2 enriched phenotype was restored. These pathways converge on a common theme of selection of a resistant phenotype, likely due to ITH.

Novel Tools for Understanding Intratumoral Heterogeneity

Novel techniques to study ITH are being developed rapidly, with examples such as single nuclei sequencing, organoid models, and three-dimensional lineage tracing. A particularly attractive technique utilized high throughput single nuclei RNA sequencing to study cryopreserved tissue, demonstrating comparable transcriptomic signatures to single cell RNA sequencing(72). Broad translational implications of this platform are apparent, as this technique allows the study of cryopreserved tissue biobanks with long term follow-up to associate clinical outcomes with transcriptomic signatures. Moreover, novel mechanisms for therapeutic resistance may be elucidated. Sachs *et al.* created a biobank of breast cancer organoids from more than 100 primary tumors and metastatic lesions(96). They found that their organoids captured the clinical features of the index lesion, including histologic and receptor status (ER/PR/HER2), as well as the mutational signatures, including mutations in known driver oncogenes. Organoids recapitulate clonal evolution, a driver of ITH, known to occur in patients. Rios *et al.* developed a large-scale single-cell three-dimensional resolution

technique, that combined with lineage tracing and clonal RNA sequencing, created a novel platform for studying ITH(97). They found clonal restriction during tumor progression with clonal EMT occurring frequently, highlighting ITH and clonal plasticity. Taken together, these novel tools promise to further our understanding of ITH, metastases, and treatment resistance.

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Future Prospects

We have come a long way in our understanding of the distinct, yet fluid phenotypes of breast cancer. Clonal heterogeneity and cell state heterogeneity create robust tumors with profound ITH, which drives disease progression and therapeutic resistance. Identifying the molecular drivers of this heterogeneity and understanding their relative contributions to metastasis and resistance will guide the development of future therapeutics (Figure 2).

One particularly interesting area of investigation is the tumor microenvironment, specifically the contribution of the microenvironment to ITH. Both pre-clinical and clinical studies have shown the niche environment can help to determine the phenotype of the malignant cells and response to therapy, as seen with periostin inhibition(83). Novel translational studies to account for tumoral heterogeneity, including longitudinal sampling with therapeutic choices determined by specific molecular aberrations, promise to improve patient outcomes. Analysis of tumors at single cell resolution (i.e., transcriptome profiling) have uncovered previously overlooked heterogeneity, hidden through bulk analysis. Three potential advances in the field of single cell analysis are a classification system for single malignant cells, streamlining single cell analytics for translational studies, and multi-omics analysis of tumors with preserved spatial relationships. A single cell classification system could allow researchers to better understand the interplay between the heterogeneous components in the mammary tumor and its microenvironment, in the same way PAM50 has advanced understanding at the bulk level, but whose reliance on proliferation genes prohibits translation to single cell analysis. Translational studies using single cell analytics in treatment naïve and treatment resistant tumors may help identify the mechanism driving increased signaling nodes in resistant clones

and thereby develop targeted therapies. Finally, analysis of single cells with preserved cytoarchitectural relationships allows us to investigate how cell-cell signaling may preserve and drive ITH. This is an area of research in its infancy, with particular need for study on the molecular drivers of spatial heterogeneity.

These novel techniques will further expand our understanding of heterogeneity in the breast tumor. Once classified into broad clinical subtypes, we now recognize that each tumor is comprised of cells with the potential for multiple distinct cell states, corresponding to a hierarchy of normal mammary development. Tumors previously classified as a single molecular subtype are now understood to contain multiple different subtypes which may interconvert through plasticity (Figure 3). It is fitting that the study of breast cancer, the first cancer whose molecular characterization into different subtypes drove rational treatment, has driven our understanding of tumor heterogeneity as we develop new strategies to develop patient specific, targeted therapy.

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944 Figure 1 – Molecular Classification Schema for Breast Cancer

945 A proposed model for understanding how molecular subtypes correspond to the initial
946 molecular schema of ER⁺/PR⁺, HER2⁺, and TNBC. The PAM50 gene signature organizes breast
947 cancers into five distinct subtypes: Luminal A, Luminal B, Normal-like, Basal, and HER2-
948 enriched, with further classification of the distinct Claudin-low subtype(2, 20). At the same
949 time, the TNBC subtype was defined into four distinct subtypes: Basal-Like 1 (BL1), Basal-Like 2
950 (BL2), Mesenchymal (M), and Luminal Androgen Receptor (LAR)(25).

951

952 Figure 2 – Causes and Consequences of Molecular Heterogeneity

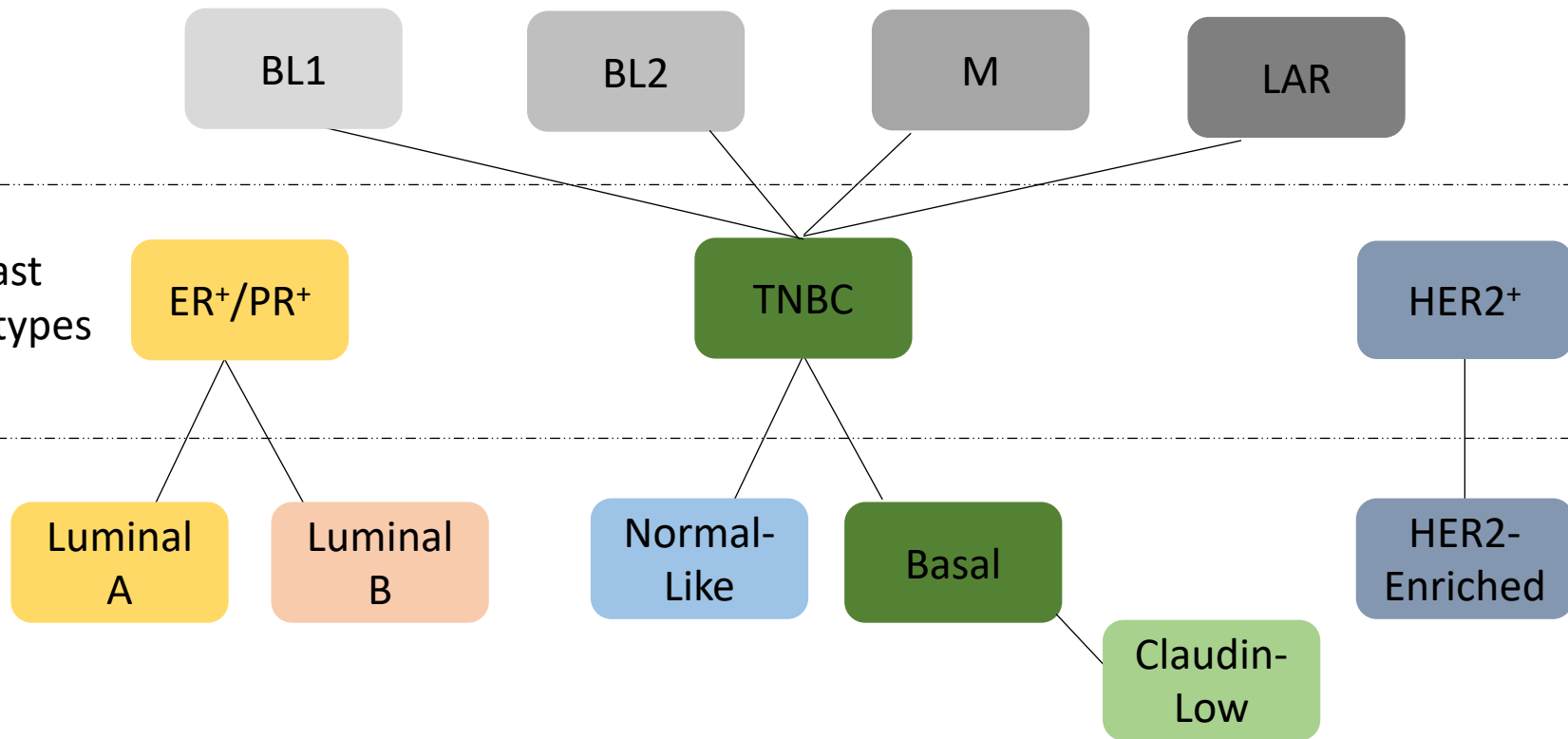
953 Listed above are potential drivers of multiple different breast cancer subtypes in a single tumor
954 and resulting effects. The formation of intratumoral heterogeneity can be caused by multiple
955 different breast cancer stem cells, dynamic conversion between subtypes, cell state transitions,
956 and tumor microenvironmental factors, among other factors. Repercussions of these
957 phenomena are the development of metastatic disease and a treatment resistant phenotype.

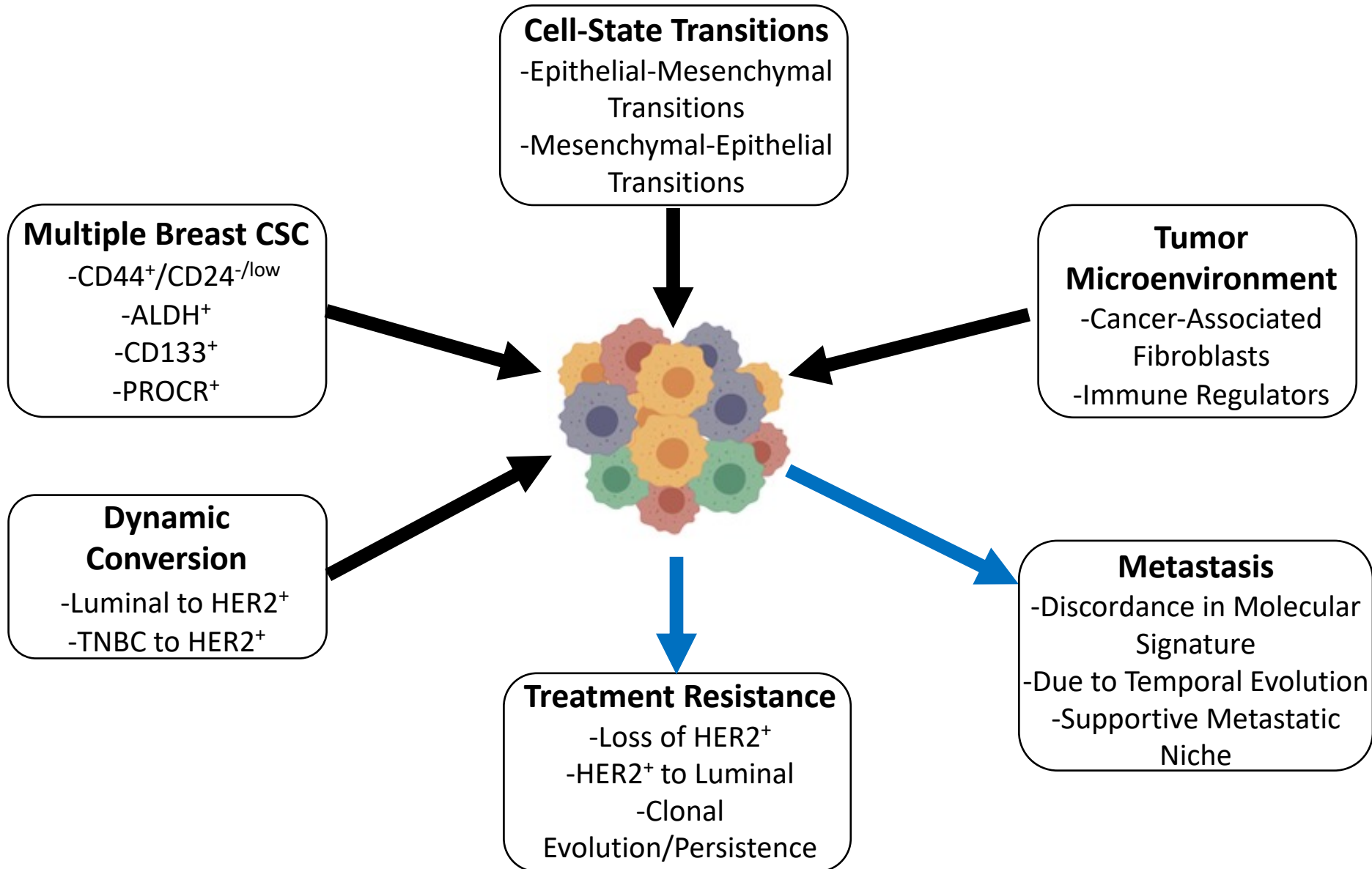
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Figure 3 – Proposed Cell State Heterogeneity Among PAM50 Subtypes

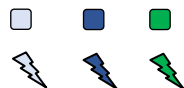
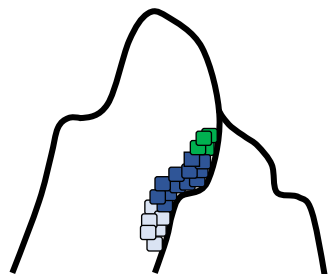
Within a single breast tumor, where at the bulk level the tumor is classified according to the dominant sub-population into the appropriate PAM50 subtype (HER2-enriched, Basal-Like, Normal-like, Luminal A, and Luminal B), there can exist cell state diversity masked at the bulk level. Herein we show a fluid model of tumor cell states corresponding to the cell states in the development of normal mammary tissue. Within each tumor subtype there can exist a plurality of cell states, with implications including the need to account for cell state heterogeneity. The presence of multiple cell state would correspond to multiple potential therapeutic targets accounting for each cell state.

Clinical Breast
Cancer Subtypes

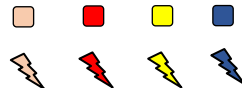
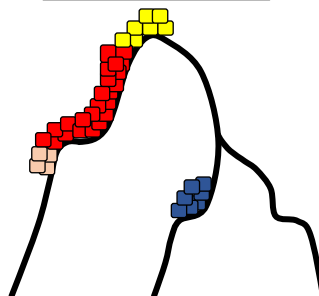




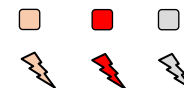
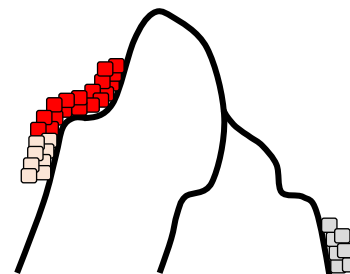
HER2 Enriched



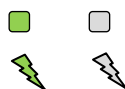
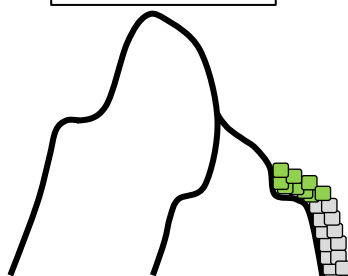
Basal



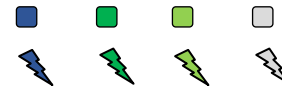
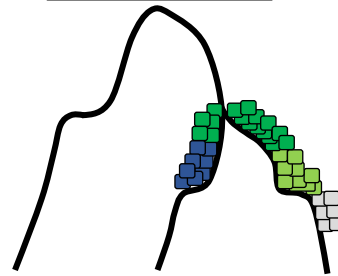
Normal-like



Luminal A



Luminal B



Potential Cell States
Therapeutic Targets

Legend:

- Bipotent Stem Cell
- Unipotent Basal Stem Cell
- Unipotent Luminal Stem Cell
- Basal Progenitor
- Alveolar Progenitor
- Hormone-Sensing Progenitor
- Basal Cell
- Alveolar Cell
- Hormone-Sensing Cell